High Level Design – Cryptocurrency Volatility Prediction

System Architecture

The system architecture of the Cryptocurrency Volatility Prediction project consists of multiple stages that process historical cryptocurrency market data and generate volatility predictions using a machine learning model.

The workflow of the system begins with loading the cryptocurrency dataset. The dataset contains historical information such as open price, high price, low price, close price, trading volume, and market capitalization.

In the next stage, data preprocessing is performed to clean the dataset. This includes removing duplicate records, handling missing values, and eliminating unnecessary columns. After preprocessing, feature engineering is applied to create new features such as daily returns, rolling mean, and volatility, which help the machine learning model understand market behavior more effectively.

Once the dataset is prepared, the processed data is divided into training and testing sets. A Random Forest Regression model is then trained using the training data to learn patterns in cryptocurrency market movements.

Finally, the trained model predicts the volatility of cryptocurrencies based on input features. The model performance is evaluated using metrics such as Mean Squared Error (MSE), and the trained model is saved for future predictions.

Architecture Flow Diagram

Cryptocurrency Dataset
↓
Data Preprocessing
↓
Feature Engineering
↓
Train-Test Split
↓
Machine Learning Model (Random Forest)
↓
Volatility Prediction
↓
Model Evaluation & Saving

Components

Data Layer
Contains the cryptocurrency dataset.

Processing Layer
Handles data cleaning and feature creation.

Model Layer
Trains the machine learning model.

Prediction Layer
Predicts volatility based on input features.